A Multi-layered Coordination Matrix Framework for Assessing Teamwork Skills in Mixed-year Student Projects

1st Ngan Nguyen Thien, 2nd Uyen Hoang Thi Thao, 3rd Van Tran Phuong Anh, 4th Anh Nguyen Ngoc Lan, 5th Chi Dao Thi Mai, 6th Nguyen Bui Tran Trong, 7th Hoa Le Thanh
Dept. of Software Engineering University of Information Technology - VNU-HCM
Ho Chi Minh City, Vietnam

Coresponding author : Truc Nguyen, Email: trucntt@uit.edu.vn

***Abstract*—Teamwork is a cornerstone of modern higher education, yet assessing individual contributions in mixed-year student projects remains a significant challenge. Traditional evaluation methods often fail to account for the disparity in experience and the value of peer mentoring between senior and junior students. This paper proposes a Multi-layered Coordination Matrix framework integrated with a Knowledge Transfer Index (KTI) to quantify both task performance and peer-to-peer knowledge sharing. A survey conducted among students across various universities in Ho Chi Minh City revealed that competence gaps significantly impact group dynamics. Results indicate that while 73.6% of students support peer assessment, there is a strong demand for a more transparent, weight-based scoring system. The proposed framework provides a scalable solution to enhance evaluation fairness and mentoring quality in project-based learning.**

***Keywords*—Teamwork skill assessment, Mixed-year projects, Multi-layered coordination matrix, Knowledge Transfer Index (KTI), Peer assessment, Project-based learning.**

I. INTRODUCTION

In the contemporary educational landscape, Project-based Learning (PBL) has emerged as a vital strategy to bridge the gap between theoretical instruction and professional practice. Within this context, mixed-year student projects—where students from different academic cohorts (e.g., freshmen to seniors) collaborate—are increasingly implemented to simulate the hierarchical and cross-functional structures of the modern workforce. This model provides a rich environment for "social learning," where senior students act as unofficial mentors, guiding juniors through complex technical tasks [1].

Despite its pedagogical potential, the mixed-year model introduces intricate group dynamics that traditional assessment methods struggle to quantify. Current evaluation systems primarily rely on final project outcomes or subjective peer-rating scales. Such methods often overlook the "invisible labor" of senior students who expend significant effort on guidance, as well as the unique progress made by junior students who may start from a lower knowledge base [2]. Consequently, this leads to a lack of perceived fairness in grading, potentially demotivating high-performing seniors and causing junior students to feel marginalized within the team structure [4].

Furthermore, research indicates a recurring "communication breakdown" in these diverse teams. Disparities in specialized knowledge often trigger psychological barriers, such as a "fear of judgment" among younger students, which can lead to a one-way flow of information rather than true collaboration. Without a transparent mechanism to track and reward proactive mentoring and individual growth, the risk of "free-riding" increases, and the quality of the collaborative experience diminishes.

This study addresses these gaps by conducting an empirical analysis of teamwork dynamics among 84 students across multiple universities in Ho Chi Minh City. We propose the Multi-layered Coordination Matrix, a structured framework designed to categorize student roles and contributions across multiple dimensions. Central to this framework is the Knowledge Transfer Index (KTI), a mathematical metric developed to quantify the efficiency of peer-to-peer mentoring. By shifting the focus from purely outcome-based metrics to behavior-oriented assessment [5], this paper provides a robust methodology for ensuring academic integrity and fostering a more equitable collaborative culture in higher education.

II. THEORETICAL FRAMEWORK

A. Group Development and Role Dynamics

The structural evolution of mixed-year teams is analyzed through the Tuckman Model of Group Development, which identifies five critical stages: Forming, Storming, Norming, Performing, and Adjourning [4]. In mixed-year contexts, the "Storming" phase is often intensified by significant disparities in technical experience and academic mindsets, which can lead to conflicts if not properly coordinated [4]. Complementing this, Belbin's Team Role Theory suggests that while seniors naturally gravitate toward "Coordinator" or "Shaper" roles, juniors typically occupy "Implementer" positions. However, this natural division can lead to a contribution imbalance if roles are not dynamically adjusted [4].

B. Social Learning and the Zone of Proximal Development

The theoretical foundation for peer mentoring in mixed-year projects is rooted in Vygotsky's Social Development Theory and Bandura's Social Learning Theory [1]. The concept of the Zone of Proximal Development (ZPD) explains how junior students bridge skill gaps through "scaffolding" provided by experienced seniors [1]. Furthermore, Collaborative Learning theories emphasize that knowledge is co-constructed through active interaction rather than passive reception. In these settings, Distributed Cognition occurs, where collective intelligence is shared across the team, enhancing the ability to solve complex technical problems.

C. *Limitations of Existing Assessment Frameworks*

Effective teamwork assessment involves dimensions such as communication, coordination, and individual contribution [1]. Current academic frameworks often utilize both Formative Assessment and Summative Assessment. However, a significant "evaluation gap" exists in mixed-year projects. Traditional tools, such as the Collaboration Skills Scale (CoSS), often rely on subjective self-reports, which are susceptible to student overestimation by approximately 0.60 standard deviations [6]. Moreover, most peer assessment instruments do not distinguish between technical output and the quality of peer-to-peer mentoring [5], necessitating a shift toward behavior-oriented frameworks.

III. Analysis of current teamwork dynamics

A. *Survey Methodology*

To establish an empirical baseline, a quantitative study was conducted via an online survey targeting students across various universities in Ho Chi Minh City. The survey analyzed 84 valid responses from participants across different academic levels. Sophomores (57.1%) and Freshmen (23.8%) represented the majority of the sample. The study utilized 5-point Likert scales to measure perceived impact, trust, and communication barriers.

B. *Empirical Results and Discussions*

The analysis of the survey data revealed several critical insights into the mixed-year project environment:

1) Impact of Competence and Experience Disparity: knowledge differences between seniors and juniors significantly impact project progress, according to 71.4% of respondents. Furthermore, 51.2% of participants noted that junior members had a lower capacity to grasp complex project requirements. This disparity necessitates a higher degree of supervision and can lead to performance bottlenecks if not managed.

2) Mentoring Dynamics and Mutual Trust: Peer mentoring is highly active, with 50% of senior students reporting that they frequently provide guidance to juniors. Despite the competence gap, 58.3% of students maintained a stable level of trust in the work performed by members of a different academic year.

3) Workload Distribution and Leadership: Factors such as actual technical competence (89.3%) and experience (83.3%) are the primary drivers of task allocation. While 72.6% of students perceive the current distribution as fair, 79.8% of team leaders are elected based on merit rather than seniority.

4) Psychological and Communication Barriers: The primary obstacle in cross-year interaction is "psychological hesitation", reported by 47.6% of students. This finding aligns with research suggesting that perceived unfairness in groupwork often stems from interactional and procedural transparency issues [2]. Additionally, 40.5% of participants cited differences in thinking styles as a barrier.

5) Evaluation and Future Expectations: Regarding grading fairness, 70.2% of students currently support equal-split scoring, yet there is a significant demand (86.1%) for weight-based systems that accurately reflect individual mentoring efforts and progress. The study also found that "Adaptability" (63.1%) is the soft skill most improved through mixed-year projects [1].

IV. Proposed method: Multi-Level Collaboration Matrix

A. *Overview of the Multilayer Collaboration Matrix*

The Multilayer Collaboration Matrix (MCM) is an integrated assessment framework specifically designed for collaborative environments involving students from diverse academic cohorts. Distinct from traditional assessment methodologies - which primarily document final summative outputs such as products or numerical grades - the MCM monitors and quantifies the entire process of interaction, coordination, and knowledge dissemination among team members throughout the project lifecycle.

The framework is established upon three fundamental theoretical pillars: Tuckman’s Group Development Model (tracking transitions through developmental stages), Belbin’s Team Role Theory (allocating responsibilities based on functional competencies), and Social Learning Theory (recording the reciprocal learning processes between participants).

B. *Matrix Structure: Role-Based Hierarchy and Contribution Tiers*

The proposed Multi-level Collaboration Matrix is designed to address the inherent hierarchy and communication gaps in mixed-year student projects. The structure is bifurcated into two primary dimensions:

1) Vertical Dimension - Role Layer: Participants are categorized into three functional groups to clarify expectations:

a) Leader/Coordinator Group: Primarily senior students responsible for orchestrating progress, task allocation, and conflict resolution.

b) Implementer Group: Comprising students from both cohorts, tasked with executing specific deliverables according to the project plan.

c) Learner/Supporter Group: Primarily junior students, focusing on knowledge acquisition, skill refinement, and auxiliary team support.

2) Horizontal Dimension - Contribution Tiers: Unlike traditional flat evaluation, this matrix assesses contributions based on the "Depth of Impact." It distinguishes between Silo Tasks (independent work) and Synergetic Tasks (cross-functional collaboration), ensuring that students who facilitate others' progress are rewarded accordingly.

Domain	Measurement Criteria	Recording Tools
Professionalism	Quality and volume of assigned technical tasks.	Work logs, Git commits, Trello task completion.
Collaboration	Frequency and quality of interaction with other members.	Meeting minutes, peer-review feedback logs.
Knowledge Transfer	Mentoring efficacy (for seniors) / Acquisition and application (for juniors).	Mentor-mentee templates, progress observation reports.
Attitude & Proactivity	Degree of self-discipline, initiative, and procedural compliance.	Periodic Pulse Surveys, cross-evaluation metrics.

C. Knowledge Transfer Index (KTI): A Quantitative Formulation

This is the core and most characteristic metric of the method, reflecting the added value that the heterogeneous grouping model provides compared to homogeneous grouping.

KTI Calculation Formula:

$$KTI = \frac{\left((w_1 * C_{mentor}) + (w_2 * P_{mentee}) + (w_3 * Q_{collab})\right)}{3}$$

Where:

- *Cmentor:* Evaluation score of the senior student's mentoring ability (assessed by junior students and instructors)
- *Pmentee:* Progress score of junior students compared to the beginning stage of the project (assessed by senior students and instructors)
- *Qcollab:* Overall collaboration quality of the mentor–mentee pair (based on direct observation and management tool logs)
- *w1, w2, w3:* Adjustable weighting factors depending on the characteristics of each project (default values: w1 = 0.35, w2 = 0.35, w3 = 0.30)

D. Implementation Roadmap: The Four-Stage Lifecycle

The operationalization of the Multi-level Collaboration Matrix follows a systematic four-step roadmap to ensure transparency and fairness:

1) Phase 1 – Foundational Setup (Week 1): The team establishes consensus on assessment criteria, assigns roles based on actual competency (rather than academic seniority defaults), and configures monitoring tools (e.g., Trello, Notion, GitHub). Instructors/mentors ratify the role distribution and criteria prior to project commencement.

2) Phase 2 – Periodic Monitoring (Continuous): On a weekly basis, members complete concise Pulse Surveys (5–7 items) regarding communication, conflict, and individual progress. Data from management tools are automatically aggregated to calculate objective contribution metrics.

3) Phase 3 – 360-Degree Evaluation (Conclusion of major milestones): Members engage in a three-tier assessment: self-evaluation, anonymous peer-assessment, and instructor/mentor feedback. These results are synthesized to determine KTI scores for each mentor-mentee pair.

4) Phase 4 – Final Synthesis and Score Calibration (Project Conclusion): The final individual score is derived as follows:

$$score_{individual} = (score_{team} * 0.5) + (KTI * 0.3) + (score_{professional} * 0.2)$$

This mechanism ensures that junior students handling smaller-scale tasks receive equitable recognition for demonstrable progress and effective collaboration, while effectively mitigating "free rider" behaviors within the group.

E. Application Conditions and Limitations

To optimize the efficacy of this methodology, several conditions must be satisfied:

1) Group Composition: Ideal size of 4–8 members, including at least one senior and one junior student.

2) Project Duration: A minimum of 4 weeks to ensure sufficient data collection for periodic evaluation cycles.

3) Member Commitment: Explicit consensus and adherence to the assessment framework from the project's inception.

4) Instructor Role: Instructors/mentors serve as arbitrators in instances of significant statistical variance in cross-evaluation results.

Limitations: Comprehensive multi-dimensional data collection necessitates high behavioral discipline from participants. Furthermore, the KTI remains a relative metric susceptible to subjective bias if anonymity is compromised. Consequently, weighting factors (w1, w2, w3) should be iteratively calibrated based on empirical feedback.

V. Discussion and Feasibility

A. Feasibility Assessment: Student Acceptance of the Mentor-Mentee Model and "Adjusted Score" System

The empirical results from the survey demonstrate substantial student endorsement for the proposed structural changes. Specifically, the Mentor-Mentee model—where senior students provide direct guidance to juniors—was rated as "highly necessary" (4 to 5 stars) by 72.2% of respondents. This support validates the application of Social Learning Theory, suggesting that junior member can significantly accelerate their skill adaptation through observational learning and direct collaboration with experienced seniors.

Furthermore, the implementation of an "Adjusted Score" mechanism to ensure grading equity received a cumulative support rate of 86.1% (48.6% fully supporting and 37.5% supporting with clear criteria). This high level of acceptance reflects a strong student demand for transparency and accountability to mitigate the "free-rider" phenomenon. These findings align with prior research at RMIT University, which indicated that individualizing scores based on peer assessment reduced "social loafing" complaints from 26% to 7% [7]. Thus, the proposed Knowledge Transfer Index (KTI) serves as a feasible and data-driven metric to quantify intangible mentoring efforts into academic credits.

B. Practical Challenges: Asynchronous Schedules and Management Tool Discipline

Despite strong theoretical support, practical implementation faces two primary hurdles: asynchronous schedules and inconsistency in management tool usage.

1) Asynchronous Learning Schedules: Variations in academic timetables remain a critical barrier to real-time collaboration. Survey data indicates that students favor "Clear task-based deadlines" as the most viable solution (73.6% approval), followed by "Fixed weekly meeting slots" (70.8%). While instant messaging platforms like Zalo or Messenger remain dominant for rapid coordination (84.7%), there is a notable resistance toward asynchronous documentation; only 25% of students found recording meetings to be an effective substitute for live interaction.

2) Discipline in Management Tools (Trello/GitHub): A significant discrepancy exists between the perceived utility and actual usage of project management platforms. Although 66.7% of students agree that tracking tasks on Trello or Notion is the most objective way to reflect individual contribution, only 31.9% maintain frequent updates. This gap suggests that the challenge is socio-technical—rooted in individual discipline rather than a lack of technology.

To mitigate these challenges, the research suggests establishing "Ground Rules" and mandatory "Onboarding Sessions" at the project's inception. Standardizing digital workflows and proactively decomposing tasks into independent modules will allow for flexible execution while maintaining the integrity of the KTI assessment.

VI. CONCLUSION

This research successfully addresses the inherent complexities of teamwork in mixed-year student projects by establishing a structured and quantitative assessment framework. The primary contribution of this study is the Multi-layered Coordination Matrix, which effectively converts intangible support activities—such as mentoring and peer guidance, often overlooked in traditional academic grading—into concrete, measurable data. By implementing the Knowledge Transfer Index (KTI), the framework provides a robust formulaic method to evaluate mentoring efficiency, the learning progression of junior members, and the quality of peer-to-peer interactions. This data-driven approach not only validates the "invisible" leadership efforts of senior students but also enhances transparency and accountability within the team. Consequently, the proposed model serves as a strategic tool to mitigate the "free-rider" effect and foster a more equitable, collaborative learning environment in higher education.

To enhance the scalability and precision of the proposed framework, future research will focus on the digitalization and automation of the assessment toolkit through several key development pathways. A primary objective is the integration of the matrix into industry-standard project management tools, such as Jira, Trello, or GitHub, to automatically capture interaction logs and communication frequency for more objective KTI calculation. Additionally, the implementation of Machine Learning algorithms and Natural Language Processing (NLP) could provide real-time, nuanced evaluations of technical discussions while reducing subjective human bias. Finally, expanding empirical testing across diverse academic disciplines—including Engineering, Economics, and Design—will be essential to refine weighting coefficients and verify the model's reliability in varied educational contexts.

ACKNOWLEDGMENT

This research was supported by The VNUHCM-University of Information Technology’s Scientific Research Support Fund.

REFERENCES

[1] M. Reque and M. Pino-Juste, "Teamwork skills in higher education: Is university training contributing to their mastery?," *Psicologia: Reflexão e Crítica*, vol. 35, no. 1, 2022.

[2] "Students' perceptions of fairness in groupwork assessment: validity evidence for peer assessment fairness instrument," *Assessment & Evaluation in Higher Education*, 2025.

[3] H. Zhao, T. N. Y. Dang, and N. Finlayson, "Operationalising critical thinking in postgraduate disciplinary writing: insights from corpus and cluster analyses of lecturer feedback," *Assessment & Evaluation in Higher Education*, 2024.

[4] D. Lee, Y. Huh, and C. M. Reigeluth, "Collaboration, intragroup conflict, and social skills in project-based learning," *Instructional Science*, vol. 43, pp. 561-590, 2015.

[5] "Closing the Evaluation Gap: Developing a Behavior-Oriented Framework for Assessing Virtual Teamwork Competency," in *ResearchGate*, 2025.

[6] "Development and Validation of the Collaboration Skills Scale (CoSS) for Formative Assessment," *Journal of Educational Research*, 2024.

[7] H.Babai, “Individualising teamwork scores based on peer assessment”, *International Journal of Educational Research Open*, 2022.